Geometry in Mathematics and Physics: selected entries

DEFINITION · math:diffgeo:geometric-structures:df:hodge-star

The Hodge star on an oriented metric manifold

Let (M, g) be an oriented pseudo-Riemannian n-manifold with @math:diffgeo:geometric-structures:df:pseudo-metric. In an oriented chart its metric volume form is

$$\mathrm{vol}_g = \sqrt{|\det(g_{ij})|}\, dx^1 \wedge \cdots \wedge dx^n.$$

The Hodge star is the unique bundle map $\star_g : \Lambda^k T^*M \to \Lambda^{n-k} T^*M$ characterized by

$$\alpha \wedge \star_g \beta = \langle \alpha, \beta \rangle_g \, \mathrm{vol}_g$$

for all k-forms α, β. The wedge pairing is nondegenerate in a basis of @math:diffgeo:forms:df:exterior-algebra, so this equation determines a unique @[smooth map]math:diffgeo:manifolds:df:smooth-map. Both metric and orientation are inputs; the @math:diffgeo:forms:df:exterior-derivative needs neither. Reversing orientation negates $\star_g$.

On oriented Euclidean three-space, $\star dx = dy \wedge dz$ and $\star(dx \wedge dy) = dz$. On Minkowski space with $x^0 = ct$, metric $(-+++)$ and orientation $dx^0 \wedge dx \wedge dy \wedge dz$, $\star(dx^0 \wedge dx) = -dy \wedge dz$ and $\star(dy \wedge dz) = dx^0 \wedge dx$. These signs fix the electromagnetic convention.

THEOREM · math:diffgeo:geometric-structures:th:hodge-square

The square of the Hodge star and conformal scaling

For the @math:diffgeo:geometric-structures:df:hodge-star on an oriented n-manifold of metric index s, every k-form satisfies

$$\star_g \star_g \alpha = (-1)^{k(n-k)+s}\alpha.$$

If $\tilde{g} = e^{2f} g$ for a smooth real function f, with the same orientation, then on k-forms

$$\star_{\tilde{g}} = e^{(n-2k)f} \star_g .$$

In four-dimensional Lorentzian signature, $\star^2 = -1$ on two-forms and their Hodge star is conformally invariant.

Proof

At one point choose an oriented orthonormal coframe with squared norms $\epsilon_i \in \{1, -1\}$, whose product is $(-1)^s$. For an increasing multi-index I, let I^c be its complement and let σ_I be the sign of the ordering (I, I^c). The defining wedge identity gives $\star e^I = (\prod_{i \in I} \epsilon_i)\sigma_I e^{I^c}$. Applying it twice contributes $(-1)^s \sigma_I \sigma_{I^c}$. Moving the k entries past the $n-k$ complementary entries gives $\sigma_I \sigma_{I^c} = (-1)^{k(n-k)}$. This proves the first formula on a basis.

For the second, the inverse metric rescales by e^{-2f}, so the pairing of k-forms rescales by e^{-2kf}, while the volume rescales by e^{nf}. Substitute both factors into the defining wedge identity and use uniqueness of the star. No derivative of f enters this pointwise identity.

THEOREM · physics:classical:geometry:th:momentum-conservation

Symmetry and conservation of the momentum map

For a cotangent-lifted action and its @physics:classical:geometry:df:moment-map, one has $\iota_{\xi_{T^*Q}}\omega = dJ_\xi$. If a smooth Hamiltonian H is invariant under the action, then J_ξ is conserved along its Hamiltonian flow for every ξ. The conclusion is local in time wherever the flow exists.

Proof

Write $\pi : T^*Q \to Q$. For a lifted map Φ_g and tangent vector W at α_q, the definition of θ gives

$$(\Phi_g^*\theta)_{\alpha_q}(W) = ((dg_q)^{-1})^*\alpha_q\Big(dg_q(d\pi(W))\Big) = \alpha_q(d\pi(W)) = \theta_{\alpha_q}(W).$$

Thus $\mathcal{L}_{\xi_{T^*Q}}\theta = 0$. Since $\iota_{\xi_{T^*Q}}\theta = J_\xi$, @math:diffgeo:forms:th:cartan and $\omega = -d\theta$ yield $\iota_{\xi_{T^*Q}}\omega = dJ_\xi$. Nondegeneracy identifies the generator with X_{J_ξ}. Invariance of H gives $dH(X_{J_\xi}) = 0$. Along its flow,

$$\frac{d}{dt}J_\xi = dJ_\xi(X_H) = \omega(X_{J_\xi}, X_H) = -dH(X_{J_\xi}) = 0.$$

This gives @physics:classical:lagrange:th:noether an intrinsic Hamiltonian form. It does not extend conservation to a force that breaks the specified symmetry.

THEOREM · physics:waves:geometry:th:optical-metric

The eikonal equation, Hamiltonian rays, and the optical metric

Let $n > 0$ be a smooth refractive index on a Euclidean domain and set the optical metric $g_{\mathrm{opt}} = n^2\delta$. Its geodesic Hamiltonian on the cotangent bundle is

$$K(x,p) = \tfrac{1}{2}|p|^2/n(x)^2.$$

If a smooth real phase S satisfies $|\nabla S|^2 = n^2$, its graph $p = dS$ lies in $K = 1/2$ and is invariant under X_K wherever it remains a graph. The projected Hamiltonian curves are unit-speed optical geodesics. Their Euclidean-arclength parametrization satisfies @physics:waves:ray-optics:th:ray-equation. The phase S has length units, and $p = dS$ has dimensionless Cartesian components.

Proof

The metric construction in @physics:classical:geometry:df:kinetic-metric, with zero potential, gives K. If s_{opt} is its Hamiltonian time parameter, then $dx/ds_{\mathrm{opt}} = p/n^2$ and on $K = 1/2$ this has optical norm one. @physics:classical:geometry:th:geometric-newton therefore gives its affine geodesic equation.

The eikonal equation is $K(x, dS) = 1/2$. Differentiating in x yields $K_x + (D^2S)K_p = 0$. The tangent to the graph along base velocity K_p is $(K_p, (D^2S)K_p)$, which equals $(K_p, -K_x) = X_K$. Hence the graph is invariant locally by uniqueness of the flow.

Write s for Euclidean arclength. The unit-optical-speed condition gives $ds_{\mathrm{opt}} = n\,ds$ and $p = n\,dx/ds$. Hamilton's second equation on $K = 1/2$ is $dp/ds_{\mathrm{opt}} = \nabla n/n$. Multiplying by n gives $d(n\,dx/ds)/ds = \nabla n$, exactly the ray equation. Conversely that equation with $|dx/ds| = 1$, the displayed p, and $ds_{\mathrm{opt}} = n\,ds$ gives both Hamilton equations.

DEFINITION · physics:electromagnetism:spacetime-forms:df:field-current-forms

Electromagnetic field, excitation, and charge-current forms

Use an oriented time-oriented Lorentzian four-manifold (M, g) with signature $(-+++)$ and @math:diffgeo:geometric-structures:df:hodge-star. The electromagnetic field is a real two-form F. The electric charge-current is a three-form $\mathcal{J}$: its integral over an oriented spacelike three-surface is the charge crossing that surface. For a four-current vector $J = c\rho\,\partial_0 + j^i\partial_i$ in an inertial SI chart (intrinsically, the vector representing the given current via the metric volume), set $\mathcal{J} = c^{-1}\iota_J \mathrm{vol}_g = c^{-1} \star J^\flat$. An excitation two-form $\mathcal{H}$ describes the constitutive response; in vacuum $\mathcal{H} = (\mu_0 c)^{-1} \star F$.

In inertial SI coordinates $x^0 = ct$ with orientation $dx^0 \wedge dx \wedge dy \wedge dz$, write $\mathrm{vol}_3 = dx \wedge dy \wedge dz$, $e = \mathbf{E}^\flat$, $b = \mathbf{B}^\flat$, $B_2 = \iota_{\mathbf{B}}\,\mathrm{vol}_3$ and $j_2 = \iota_{\mathbf{j}}\,\mathrm{vol}_3$. The flats in this spatial sentence use the Euclidean metric. Then

$$F = B_2 - dt \wedge e, \qquad \mathcal{J} = \rho\,\mathrm{vol}_3 - dt \wedge j_2,$$

$$\star F = c^{-1} \star_3 e + c\,dt \wedge b, \qquad \mathcal{H} = \epsilon_0 \star_3 e + \mu_0^{-1} dt \wedge b.$$

Thus $F_{0i} = -E_i/c$ and $F_{ij} = \epsilon_{ijk} B_k$, exactly as in @physics:relativity:covariant-fields:df:field-tensor. Integrals of F have magnetic-flux units and integrals of $\mathcal{H}$ have charge units. Orientation is fixed here; formulations on nonorientable manifolds use orientation-twisted forms and are outside this entry.

THEOREM · physics:electromagnetism:spacetime-forms:th:maxwell-forms

Maxwell equations as exterior differential equations

With the smooth fields and conventions of @physics:electromagnetism:spacetime-forms:df:field-current-forms on Minkowski spacetime, the four vacuum Maxwell equations with electric sources are equivalent to

$$dF = 0, \qquad d\mathcal{H} = \mathcal{J}, \qquad \mathcal{H} = \frac{1}{\mu_0 c} \star F.$$

Equivalently $d \star F = \mu_0 c\, \mathcal{J} = \mu_0 \star J^\flat$. The same intrinsic equations define the minimally coupled Maxwell model on a curved oriented Lorentzian spacetime; that extension is a physical modeling choice. In components it is $\nabla_\nu F^{\mu\nu} = \mu_0 J^\mu$ together with $\nabla_{[\alpha} F_{\beta\gamma]} = 0$.

> ### Proof
>
> Write $d = d_3 + dt \wedge \partial_t$ on time-dependent spatial forms. The graded product rule gives
>
> $$dF = d_3 B_2 + dt \wedge (\partial_t B_2 + d_3 e).$$
>
> Here $d_3 B_2 = (\nabla \cdot \mathbf{B})\,\mathrm{vol}_3$ and $d_3 e = \iota_{\nabla \times \mathbf{E}}\,\mathrm{vol}_3$. Its two independent spatial/time components are exactly the magnetic Gauss and Faraday equations.
>
> Similarly,
>
> $$d\mathcal{H} = \epsilon_0 (\nabla \cdot \mathbf{E})\,\mathrm{vol}_3 + dt \wedge \Big(\epsilon_0 \partial_t \star_3 e - \mu_0^{-1} d_3 b\Big).$$
>
> Equating its components with $\mathcal{J}$ gives $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$ and $\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \mu_0 \epsilon_0 \partial_t \mathbf{E}$. Each decomposition is unique, proving both directions. The vector identities used here follow by expanding the @math:diffgeo:forms:df:exterior-derivative in an oriented Cartesian coframe.
>
> For the stated curved model, at any point choose oriented normal coordinates. First derivatives of g vanish there, so the same pointwise calculation gives the component equations with covariant derivatives. Both sides are tensors, so equality in that chart at each point proves their coordinate-independent equivalence. This does not derive the curved physical model from the flat one.

THEOREM · physics:electromagnetism:spacetime-forms:th:maxwell-action

The differential-form Maxwell action and gauge invariance

Let A be a real one-form on a fixed oriented Lorentzian four-manifold, $F = dA$, and let the prescribed current three-form $\mathcal{J}$ be closed. With @[compactly supported]math:diffgeo:manifolds:df:partition-unity variations, the SI action

$$S[A] = -\frac{1}{2\mu_0 c}\int_M F \wedge \star F + \int_M A \wedge \mathcal{J}$$

is stationary exactly when $d \star F = \mu_0 c \mathcal{J}$. Gauge changes $A \mapsto A + d\chi$ with a smooth compactly supported function χ leave the action unchanged. Integrals must exist; alternatively define the action on a compact region with variations supported in its interior.

> ### Proof
>
> At fixed metric, symmetry of the form pairing gives $\delta(F \wedge \star F) = 2d(\delta A) \wedge \star F$. By the graded product rule,
>
> $$d(\delta A \wedge \star F) = d(\delta A) \wedge \star F - \delta A \wedge d \star F.$$
>
> Integrate and use @math:diffgeo:stokes-cohomology:th:stokes with @[compact support]math:diffgeo:manifolds:df:partition-unity. Thus $\delta S = \int \delta A \wedge (\mathcal{J} - (\mu_0 c)^{-1} d \star F)$. Testing arbitrary compactly supported one-forms in a chart forces each complementary three-form coefficient to vanish; conversely that equation makes every first variation zero.
>
> The field term is gauge invariant because $d^2 = 0$. The source term changes by $\int d\chi \wedge \mathcal{J} = \int d(\chi \mathcal{J}) - \int \chi d\mathcal{J} = 0$. Boundary gauge transformations can leave a boundary term and are not included in this compact-support assertion. In the SI split, the source term is $\int (-\rho\phi + \mathbf{j} \cdot \mathbf{A})\, dt\, d^3x$, matching @physics:relativity:covariant-fields:rk:field-action.

THEOREM · physics:thermo:geometry:th:legendre-contact

Legendre changes preserve the thermodynamic contact form

For the contact coordinates in @physics:thermo:geometry:df:thermodynamic-contact, replacing U by $F = U - TS$ and the pair (S, T) by $(T, -S)$ preserves the contact form:

$$\alpha = dF + S\,dT + p\,dV - \mu\,dN.$$

More generally, for $\alpha = dz - \sum_i y_i\,dx^i$ and any subset I of indices, set

$$\tilde{z} = z - \sum_{i \in I} x^i y_i, \qquad \tilde{x}^i = y_i, \quad \tilde{y}_i = -x^i \quad (i \in I),$$

leaving other pairs unchanged. Then $d\tilde{z} - \sum_i \tilde{y}_i\,d\tilde{x}^i = \alpha$. On an equilibrium graph, expressing the transformed potential as a smooth function of the new independent variables requires the corresponding partial Legendre map to be locally invertible.

Proof

Differentiate $\tilde{z}$. For each exchanged pair, its contribution is $-y_i dx^i - x^i dy_i$. Subtracting $\tilde{y}_i\,d\tilde{x}^i = -x^i dy_i$ cancels the second term and leaves $-y_i dx^i$. The unchanged pairs already have that form. This proves the ambient identity and includes the Helmholtz case by direct substitution.

The transformation of all ambient coordinates is a @[diffeomorphism]math:diffgeo:manifolds:df:smooth-map, with inverse $x^i = -\tilde{y}_i$, $y_i = \tilde{x}^i$, $z = \tilde{z} - \sum_{i \in I} \tilde{x}^i \tilde{y}_i$. Restriction to an equilibrium graph is a separate coordinate question: one must solve its equations of state for the exchanged extensive variables. The @math:uganalysis:metric-multivariable:th:inverse-function theorem gives a local solution when the relevant Jacobian is invertible. The contact identity does not guarantee that condition or global single-valuedness across phases.

THEOREM · physics:statistical:geometry:th:fisher-covariance

The Hessian of log partition is a covariance metric

For @physics:statistical:geometry:df:exponential-family,

$$\partial_a \Psi = \langle A_a \rangle, \qquad G_{ab} = \partial_a \partial_b \Psi = \langle (A_a - \langle A_a \rangle)(A_b - \langle A_b \rangle) \rangle.$$

The matrix is positive semidefinite, and @[positive definite]math:algebra:linear-foundations:df:bilinear-forms at every finite parameter exactly when the family is minimal. In a minimal family the mean coordinates $u_a = \partial_a \Psi$ are valid local coordinates. The local Legendre dual $\Psi^*(u) = \eta^a u_a - \Psi(\eta)$ satisfies $D^2 \Psi^* = G^{-1}$ in those coordinates.

Proof

Differentiate the finite normalizing sum to obtain $\partial_a \Psi = \sum_i p_i A_a(i)$. Consequently $s_a = A_a - \langle A_a \rangle$ and $\partial_b p_i = p_i s_b$. Differentiating the mean once more gives the covariance formula and the definition of G.

For a real parameter vector c, $c^a G_{ab} c^b$ is the variance of $\sum_a c^a A_a$. Every p_i is positive, so this variance vanishes exactly when that observable is constant on all states. This proves both semidefiniteness and the equivalence with minimality. In the minimal case the derivative of $\eta \mapsto u$ is invertible, so @math:uganalysis:metric-multivariable:th:inverse-function supplies a local inverse. Differentiating Ψ^* cancels $u_a\, d\eta^a - d\Psi$, leaving $d\Psi^* = \eta^a du_a$. Differentiating again gives $D_u \eta = (D_\eta u)^{-1} = G^{-1}$. No global coordinate claim is needed.

THEOREM · physics:relativity:observer-geometry:th:observer-reconstruction

Observer splitting reconstructs the electromagnetic two-form

With @physics:relativity:observer-geometry:df:observer-fields, the field is uniquely reconstructed by

$$F = c^{-1} u^\flat \wedge e_u + \star(u^\flat \wedge b_u).$$

For a charge with four-velocity V the Lorentz-force covector is $f^\flat = -q\, \iota_V F$, so $g(f, V) = 0$. The force law is a physical model; its orthogonality follows algebraically from antisymmetry.

> ### Proof
>
> Choose an oriented orthonormal coframe at the event with $u = \partial_0$, $u^\flat = -dx^0$. Any two-form uniquely splits into $dx^0 \wedge \alpha + \beta$ with α spatial of degree one and β spatial of degree two. Contraction gives $\alpha = -e_u/c$. With the chosen Lorentzian star, $\star(-dx^0 \wedge b_u) = \star_3 b_u$; contraction of its four-dimensional star recovers b_u, using $\star^2 = -1$ on two-forms from @math:diffgeo:geometric-structures:th:hodge-square. Thus $\beta = \star_3 b_u$, proving existence and uniqueness. The basis calculation is tensorial and hence valid in every chart.
>
> For any vector W, $(-q\iota_V F)(W) = -qF(V, W) = qF(W, V)$, which has components $qF_{\mu\nu}V^\nu$. Raising the first index yields the convention in @physics:relativity:covariant-fields:th:covariant-lorentz-force. Evaluating on V gives zero since $F(V, V) = 0$.

PROBLEM · physics:quantum:geometry:pb:spin-berry-sphere

The Bloch sphere metric and a spin Berry phase

Consider the normalized spinor

$$\psi(\theta, \phi) = \begin{pmatrix} \cos(\theta/2) \\ e^{i\phi} \sin(\theta/2) \end{pmatrix}, \qquad 0 < \theta < \pi.$$

It is the $+1$ eigenvector of $\mathbf{n} \cdot \boldsymbol{\sigma}$, where $\mathbf{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. Compute its Fubini–Study metric, Berry connection and curvature. Find the holonomy around the latitude $\theta = \theta_0$ with ϕ increasing from 0 to 2π. State what extra physical condition is needed to identify this with the geometric phase of an evolving eigenstate.

Solution

Differentiation gives $\langle\psi, \partial_\theta\psi\rangle = 0$ and $\langle\psi, \partial_\phi\psi\rangle = i\sin^2(\theta/2)$. Subtracting the vertical part in @physics:quantum:geometry:df:projective-state-space gives

$$ds^2_{\mathrm{FS}} = \tfrac{1}{4}\left(d\theta^2 + \sin^2\theta\, d\phi^2\right).$$

Thus our metric is the round-sphere metric of radius $1/2$ in these dimensionless coordinates. The connection and curvature are

$$\mathcal{A} = -\tfrac{1}{2}(1 - \cos\theta)\, d\phi, \qquad \mathcal{B} = -\tfrac{1}{2}\sin\theta\, d\theta \wedge d\phi.$$

Consequently $\gamma = -\pi(1 - \cos\theta_0)$ modulo 2π, and the holonomy is $e^{i\gamma}$. The phase is minus one half the oriented solid angle of the north polar cap. The displayed unit frame is regular near the north pole in a suitable Cartesian chart but requires a different gauge at the south pole; the global curvature is regular.

For a Hamiltonian such as $H(\lambda) = -\Delta\mathbf{n}(\lambda) \cdot \boldsymbol{\sigma}/2$ with fixed $\Delta > 0$, this line is its nondegenerate ground eigenspace. Connecting holonomy to physical evolution requires an adiabatic regime that suppresses transitions across the gap Δ, and separation of the dynamical phase. Merely specifying the loop does not establish that regime. The sign also depends on the chosen eigenline and traversal orientation.

THEOREM · physics:continuum:geometry:th:kelvin-circulation

Kelvin circulation and transported vorticity

Suppose smooth inviscid flow with $\rho > 0$ on a fixed oriented @[Riemannian manifold]math:diffgeo:connections:df:riemannian-metric satisfies

$$\partial_t v + \nabla_v v = -\operatorname{grad}_g(h(\rho) + \Phi),$$

where $\Phi(x,t)$ is a smooth potential and $h'(\rho) = p'(\rho)/\rho$ for a barotropic pressure law $p = p(\rho)$. Then, with $u = v^\flat$ and $\Omega = du$ from @physics:continuum:geometry:df:material-forms,

$$(\partial_t + \mathcal{L}_v)u = -d\Big(h + \Phi - \tfrac{1}{2}|v|_g^2\Big), \qquad (\partial_t + \mathcal{L}_v)\Omega = 0.$$

The circulation $\oint_{\Phi_t(C)} u_t$ of every transported smooth closed curve is constant. Smoothness, positive density, a barotropic pressure law and a globally defined force potential on the region considered are hypotheses; shocks and arbitrary viscous or nonbarotropic forces are not covered.

Proof

Metric compatibility and zero torsion imply, for any vector field w,

$$du(v,w) = g(\nabla_v v, w) - g(\nabla_w v, v) = g(\nabla_v v, w) - d(\tfrac{1}{2}|v|_g^2)(w).$$

Consequently $(\nabla_v v)^\flat = \iota_v du + d(|v|_g^2/2)$. Lower the index in the assumed Euler equation and use $\mathcal{L}_v u = \iota_v du + d(u(v))$ from @math:diffgeo:forms:th:cartan, with $u(v) = |v|_g^2$. This gives the displayed one-form equation, including its minus sign before kinetic energy on the right.

The exterior derivative commutes with ∂_t and with $\mathcal{L}_v$; the latter follows immediately from Cartan's formula and $d^2 = 0$. Applying d therefore gives advection of Ω. Finally @math:diffgeo:geometric-structures:th:transport-form differentiates circulation along a material curve. Its derivative is the integral of the exact one-form on the right, which vanishes on a closed curve by the fundamental theorem of calculus, or by @math:diffgeo:stokes-cohomology:th:stokes on a one-dimensional domain with empty boundary. The curve need not bound a surface.

THEOREM · physics:gravitation:intrinsic-fields:th:killing-current

A Killing symmetry gives a closed stress-energy current

Let (M, g) be an oriented Lorentzian spacetime, T^{ab} a smooth symmetric tensor with $\nabla_a T^{ab} = 0$, and K a @math:diffgeo:geometric-structures:df:killing. Then $j^a = T^{ab} K_b$ has zero divergence, and

$$\mathcal{C}_K = \iota_j \operatorname{vol}_g, \qquad d\mathcal{C}_K = 0.$$

Its flux is conserved between compact spacelike sections with a compact connecting region and no side-boundary flux. The sign of a measured charge additionally depends on the chosen surface orientation. A general spacetime need not possess a timelike Killing field, so this does not provide a universal global energy.

Proof

Metric compatibility and the product rule give $\nabla_a j^a = (\nabla_a T^{ab}) K_b + T^{ab} \nabla_a K_b$. The first term is zero; symmetry of T reduces the second to $\frac{1}{2} T^{ab} (\nabla_a K_b + \nabla_b K_a) = 0$. In oriented coordinates, $\nabla_a j^a = |\det g|^{-1/2} \partial_a (|\det g|^{1/2} j^a)$, using $\Gamma^b_{ba} = \partial_a \log \sqrt{|\det g|}$. This is the volume divergence. Apply @physics:foundations:geometric-models:th:conserved-current-form and its boundary statement.

THEOREM · physics:fields:geometric-field-theory:th:yang-mills-variation

Variation of the source-free Yang-Mills action

On an oriented pseudo-Riemannian manifold of dimension $n \geq 2$, let A be a connection for a matrix Lie group G with a fixed symmetric nondegenerate Lie-algebra pairing B invariant under the whole group: $B(\mathrm{Ad}_g\, a, \mathrm{Ad}_g\, b) = B(a, b)$ for every $g \in G$, where $\mathrm{Ad}_g\, a = gag^{-1}$. Differentiation gives $B([a, b], c) + B(b, [a, c]) = 0$. This infinitesimal identity alone would not guarantee invariance under disconnected components of G. Extend B to algebra-valued forms by wedging their form factors. Let $F = dA + A \wedge A$ locally and fix g. For a nonzero constant κ, the action

$$S[A] = -\frac{1}{2\kappa} \int_M B(F \wedge \star F)$$

has Euler-Lagrange equation $d_A \star F = 0$ for arbitrary smooth @[compactly supported]math:diffgeo:manifolds:df:partition-unity connection variations. Independently, every connection obeys $d_A F = 0$. The statement assumes a smooth classical field and existing integrals, or a compact region with interior-supported variations.

> ### Proof
>
> A variation $a = \delta A$ is a Lie-algebra-valued one-form (globally in the bundle whose local fibers are the @[Lie algebra]math:diffgeo:geometric-structures:df:lie-group with conjugation as frame changes). Differentiating F gives $\delta F = da + A \wedge a + a \wedge A = d_A a$. Symmetry of the metric pairing yields $\delta S = -\kappa^{-1} \int B(d_A a \wedge \star F)$. Adjoint invariance cancels connection-commutator terms in the graded product rule:
>
> $$dB(a \wedge \star F) = B(d_A a \wedge \star F) - B(a \wedge d_A \star F).$$
>
> Stokes and @[compact support]math:diffgeo:manifolds:df:partition-unity therefore give $\delta S = -\kappa^{-1} \int B(a \wedge d_A \star F)$. Nondegeneracy of B and arbitrary test forms force $d_A \star F = 0$, and this equation conversely makes the variation zero. The remaining equation is @math:diffgeo:geometric-structures:th:bianchi. Under a frame change F and $d_A \star F$ conjugate, so adjoint invariance makes the action and equation globally meaningful. In the abelian case commutators vanish and this reduces to the vacuum form Maxwell equation, with physical coupling and units chosen separately.